IRSpectra-Bench and IRexp: candidate recall, not verification, limits LLM elucidation from real experimental IR and NMR

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We release *IRexp*, the largest openly redistributable collection of experimental infrared band lists (121,233 records; 43,060 structure-linked; 33,201 full IR + ^{1}H + ^{13}C + structure quadruples), and *IRSpectra-Bench*, a blind mechanically scored benchmark of 194 compounds drawn from it. On IRSpectra-Bench, given the molecular formula and peak lists exactly as reported in open-access papers, a frontier large language model recovers the correct constitution for 28% (top-1; 95% CI 22–35), or 15% once reweighted to the corpus composition — far below the near-100% implied by curated demonstrations. The bottleneck is candidate proposal, not verification: the true structure enters the pool for only 34% of compounds, and where it does, training-free forward-verification selects it 89% of the time (58/65; 81% on the 37 compounds where a ranker had a choice). Recovered from published top-*k* figures, the same split carries 68–83% of the accuracy collapse when three systems move from curated or simulated data to real heterogeneous spectra. Generating wider lifts recall 32% → 42% and top-1 23% → 30% on 60 compounds; verification itself moves whole-benchmark top-1 only 28% → 30%. Masking the spectra drops top-1 from 23% to 5%, and Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol all verify better than they generate. All predictions and code are released.

0.1 1. Introduction

Determining structure from spectra is central to synthetic and analytical chemistry. Machine learning uses encoder–decoder models on paired corpora; *Spectro*, for one, learns ^{1}H/^{13}C/IR → SELFIES from 6,833 molecules[1]. General-purpose LLMs are now reported off-the-shelf: a non-peer-reviewed 2026 industrial white paper found Claude Opus matching or beating commercial NMR-prediction software forward (structure → spectrum, ±0.08 ppm ^{1}H) and "recovering all eight simpler structures on every attempt" in the inverse[2]. We treat those numbers as a claim to test, not an established baseline.

That evaluation is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, seven given the *starting-material structure* as a hint, and "recovery" scored leniently over three runs and three ranked candidates. The chemist's question — *take an arbitrary experimental spectrum from a paper and recover the structure* — remains open. It needs open experimental data at scale, a blind benchmark others can report against, and honest accounting of which stage of elucidation binds.

We provide all three. *IRexp* (§2) is, to our knowledge, the largest openly *redistributable* collection of experimental IR *band lists* (121,233 records; 43,060 structure-linked; 33,201 IR + ^{1}H + ^{13}C + structure); view-only libraries such as SDBS hold more structure-linked *spectra* but cannot be redistributed (§2.1). *IRSpectra-Bench* (§3) is the blind, mechanically scored benchmark built from it — 194 compounds, complexity-stratified, multimodal IR + ^{1}H + ^{13}C — not the first such suite (MolQuest scores 530 post-2025 compounds by exact canonical SMILES[3]) but, to our knowledge, the first openly redistributable one on literature-reported peak lists with a recall/verification decomposition measured on Claude, replicated across three other model families (§4.7), with a within-compound solver-methodology control (§4). Ground truth is literature structures resolved deterministically (OPSIN/RDKit) and checked mechanically (560/560 bands on a seed-fixed sample, §2.3; expert-chemist review prepared but not yet run, §8); no LLM curates labels or scores predictions.

Forward-verification elucidation (§5) turns the model's *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry); the loop is prior art (§1.1), the decomposition ours, run training-free on every compound. The finding is sharp asymmetry: given a candidate set containing it, the model *verifies* the correct structure 89% of the time yet *proposes* it for only 34%, so generation binds the result (Fig. 1). Gain is bounded (top-1 28% → 30% on n=194; 23% → 30% on the 60-compound arm where wide generation was also tested) and, by our own measurements, cannot exceed a recall/precision ceiling without sharper verification or 2D-NMR data.

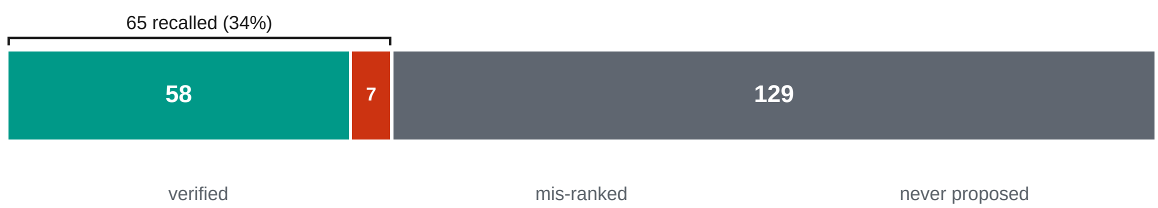

Fig. 1. Diagnosis on IRSpectra-Bench (n=194): generation recall, not verification, is the bottleneck. Where the true structure is proposed, forward-verification ranks it first 89% of the time (58/65); where it is never proposed (66% of the bar), no ranker can recover it. End-to-end top-1 is the product of those two rates (≈30%), which have different denominators and are not differenced.

Solver and verifier (Fig. S1) are LLM agents on a consumer subscription, with no fine-tuning and no API spend. Inference is not exactly reproducible — no pinned snapshot, temperature or seed (§9) — but scoring is: predictions, ground truth and scorers are released, and every training-free number regenerates from them. The HOSE lookup, GNN (§5.4) and trained generator (§5.6) are fenced complements.

0.1.1 1.1 Related work

Trained spectra → structure models. The dominant line trains sequence or graph decoders on paired corpora — *Spectro*[1], multitask 1D-NMR models[4], NMRTrans/NMRSpec[5], *NMIRacle* (IR + ^{1}H + ^{13}C)[6] — accurate in-distribution but retrained per modality. IRexp (§2.3) complements NMRSpec with redistributable IR band lists; neither corpus alone is multimodal. Upstream IR work from our group (*vIR-OLO*[7], *j-IR-vis*[8]) identifies functional groups but does not generate candidate structures.

LLMs as elucidators. LLM agents already plan syntheses (ChemCrow[9], Coscientist[10]) and call elucidation routines; we measure what one returns. Off-the-shelf and multimodal LLMs[2,11–13], puzzle benchmarks (MolPuzzle[14]), and agentic IR interpreters (IR-Agent[15]; Priessner *et al.*[16]) improve *how* a model reads spectra or re-ranks a fixed pool; we ask which stage binds once it has read them.

Most benchmarks use simulated or single-instrument spectra[1,6,17,18]; IRSpectra-Bench uses literature-reported peak lists across thousands of laboratories. Recent suites bound the task elsewhere (NMRGym[19], MolQuest[3], SpecX[20]). Cross-paper numbers swing with harness: GPT-4o scores 1.4% on MolPuzzle[14], 27.8% with chain-of-thought, 57.8% with tree-search[13]. We fix one pre-registered RDKit-InChIKey protocol with bootstrap CIs.

Forward verification and CASE. Matching predicted to observed shifts — DP4[21,22], NMR crystallography[23,24], CASE[25,26] — is easier than inverse generation. We swap DFT for a forward LLM. CASE enumerators have exhaustive recall by construction; the LLM literature almost never reports whether the true structure was proposed at all (§6).

Prior loops and concurrent work. *NMR-Solver*[27] runs the same generate-and-verify loop without an LLM (52.9% top-1 on literature ^{1}H/^{13}C, formula supplied; 31.6% with stereochemistry). We decompose error rather than beat that score. *NMRAgent*[28] is a complementary best-effort system. Kliuev *et al.* place the best of six LLMs at 24% on 105 peak lists[29]. Espejo Morales *et al.* frame agentic search with processing tools and hard checks: 80.9% on education spectra, 20.6% on 34 industrial samples[30] — they ask which architecture; we ask which stage binds. Their inputs are raw instrument files; ours are published reports (§8). Priessner *et al.* already state that re-ranking cannot recover a missing candidate[16]. We add measurement at scale: the split on 194 compounds across four model families and four verifiers, recovered from published top-k figures (§6).

0.2 2. The IRexp dataset

0.2.1 2.1 Motivation

An IRexp record is a *band list* — peak positions in cm^{-1} transcribed by an author into a paper's text — with co-reported ^{1}H/^{13}C shift lists and, where resolvable, the 2D structure. IRexp contains no absorbance traces. That is the published form a language model consumes, but a different object from a digitised spectrum, and the two should not be counted against one another.

Among *digitised spectra*, free downloads run to the NIST WebBook[31] (≈16k) and the Chemotion ELN deposit[32] (≈2k). AIST SDBS[33] is larger (≈54k FT-IR, all structure-linked) but *view-only* (50 spectra/day, no bulk export), so IRexp does not compete there: SDBS alone holds more structure-linked spectra than IRexp holds structure-linked band lists. Among text-derived *band lists* IRexp is the largest *openly redistributable* collection by record count — deliberately scoped to that object type.

0.2.2 2.2 Construction

Experimental sections report per-compound IR band lists with co-reported ^{1}H/^{13}C NMR in a stable textual convention. Mining it is mature[34]; we add a pipeline specialised to the *IR band list*, the modality those tools passed over.

- **Discovery.** Open-access literature is harvested in bulk from the PMC Open-Access Subset[35] and the Chemotion FT-IR deposit (RADAR4Chem).
- **Extraction.** A deterministic parser segments experimental text per compound and extracts IR wavenumbers and ^{1}H/^{13}C shifts, gated against scan-range artefacts and prose false-positives.

- **Resolution.** IUPAC names go to SMILES with OPSIN[36] and RDKit[37] canonicalisation (InChIKey, SELFIES[38]), with a PubChem[39] fallback for trivial names; handling open-access label conventions and narrative artefacts raised structure coverage from 24% to 35%.

0.2.3 2.3 Contents and licensing

Table 1 summarises contents and provenance.

Table 1. IRexp dataset contents and provenance.

field	value
experimental IR band-list records	**121,233**
…co-reporting ^{1}H and/or ^{13}C NMR	87,075 (72%)
…with a resolved 2D structure	43,060 (35.5%)
…of these, with ^{1}H and/or ^{13}C NMR	40,702
…of these, with both ^{1}H and ^{13}C (full quadruples)	33,201
by provenance: author-transcribed from PMC-OA text (CC-BY)	119,345
by provenance: peak-picked from Chemotion ELN deposits (CC-BY-SA)	1,888

The pools differ: PMC records are author-transcribed (median 9 bands), Chemotion records peak-picked from deposited spectra (median 39); users training on band density should treat them apart. `scripts/split_license_pools.py` separates them on `source_doi` and stamps each record's licence.

Extraction fidelity is measured. On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`) we re-fetched each article and checked every recorded wavenumber against its text: 560/560 bands and 60/60 records were confirmed (Wilson 95% CI 99.3–100% and 94–100%). This bounds *transcription* error — hallucinated, mis-parsed or unit-mangled values — below 1% of bands. It does not measure whether the parser found every IR string in every paper — a recall question requiring human reading; that audit is prepared but not yet run (§8).

`irexp_resolved` (43,060 records, 100% structure-linked) is the benchmark-ready split, ≈6× the 6,833-molecule set used to train Spectro[1] (Fig. S2).

0.3 3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw *IRSpectra-Bench*, 194 blind elucidation problems, each giving the *molecular formula* (as from HRMS), the *IR band list* and the *^{1}H and ^{13}C shift lists*; no name, SMILES or scaffold hint. In 10 of the 194 the authors' peak assignments name a ring system (`2CH2·pyrrolidine`, `pyridazinone H5`). Those compounds are solved 1/10 against 54/184 (29.3%) for the rest (`scripts/prompt_leakage.py`), so the annotation does not carry the headline; we keep them rather than drop them. Main-round ground truths are spectrally validated by an automated RDKit check (^{13}C peaks vs symmetry-unique carbons, formula match, SELFIES round-trip) excluding merged or incomplete spectra — 6/140, leaving 134; `scripts/validate_benchmark.py` applies the same filter to all three rounds, regenerating every `clean_qids.json` from the released questions and answers.

The filter does not gate on ^{1}H. In 13 of the 194 retained records the reported ^{1}H integral exceeds the reference structure's hydrogen count — residual solvent, water, exchangeable protons, or a rotamer mixture — printed as a diagnostic, not excluded; the cohort was fixed in advance, and re-filtering post hoc on a criterion chosen after seeing results is a degree of freedom a benchmark should not take. We report the sensitivity instead: dropping all 13 moves the headline from 28.4% to 29.3% (53/181), +0.9 points, leaving every conclusion unchanged. The controlled rounds' 60 compounds are fixed, pre-registered sets, audited the same way but reported rather than filtered (57/60 pass; §8).

Difficulty is declared in advance, as a property of the compound. By RDKit ring analysis a compound is *simple* iff it has at most two rings, no fused/spiro/bridgehead system and $\leq$22 heavy atoms; all others are *complex* (98 simple / 96 complex). The threshold is not load-bearing: sweeping it from 18 to 26 leaves the simple-minus-complex top-1 gap inside a 36–40-point band throughout (36.1 at the narrowest, 39.6 as released; `scripts/difficulty_sensitivity.py`). This axis is distinct from the continuous size gradient of §4.1 ($\leq$15 / 16–25 / >25 heavy atoms), and InChIKey de-duplication across rounds prevents leakage.

The 50/50 balance is by design. The sampler fills each difficulty stratum to half the round; the eligible corpus is 17.5% simple / 82.5% complex. Reweighted to that composition, top-1 falls 28.4% $\rightarrow$ 15.2% [10.6–20.4] and recall 33.5% $\rightarrow$ 19.8% [14.4–25.8]; verification precision must be reweighted too (89.2% $\rightarrow$ 71.5% [50.2–92.5]). We report the benchmark figure as released and the reweighted figure as the better estimate on an arbitrary paper.

Scoring is mechanical. A prediction is *correct* if its RDKit InChIKey connectivity layer (first 14 characters) matches the reference: we score *constitution*, so correct constitution with wrong stereochemistry counts as correct. The *full*, stereochemistry-sensitive InChIKey gives 21.1% top-1 and 25.8% recovered (41/194 and 50/194) against 28.4% / 33.5% — 7.3 points lower on top-1 and 7.7 on recovered — so the constitution figure is an upper bound on full-stereochemistry accuracy. Constitution is the headline because 1D ^{1}H/^{13}C/IR rarely fixes absolute configuration: only 10.3% (20/194) of answers carry a *defined* (assigned R/S) stereocentre, and a model is penalised there for information the prompt never contained (`scripts/score_main.py --stereo`).

Metrics. *Top-1 (exact constitution)*: best-ranked candidate matches at the connectivity layer; *recovered (top-3)*: reference among up to three returned candidates, the lenient "recovery" protocol of ref.[2]. *Generation recall*: reference in the candidate pool *before* re-ranking, the ceiling any verifier can reach (coincides with recovered except in §5.3, where the pool is larger). *Verification precision (conditional on recall)* over recall-positive compounds isolates verification from generation. The forward-verifier ranks by symmetric *chamfer distance* between predicted and observed ^{13}C peak sets (lower is better); Morgan(2, 2048) Tanimoto[40] gives a graded scaffold-family signal. CIs are bootstrap 95%; model-vs-model differences use McNemar's exact test[41] with Holm correction[42].

Solvers are frontier LLM agents (Claude Opus), one sub-agent per batch, closed-book under a consumer subscription: no tools beyond an RDKit formula check, no ground truth, verified by grep-auditing transcripts.

Formula adherence is high but imperfect. Of the 126 candidates carried through forward-verification on the original arm (§5.2), 91.3% (115/126) match the given formula exactly, and 76.6% across the full top-3 pool. On top-1 answers adherence is 95.0% (38/40) in v3 and 90.0% (18/20) in the v2-control — the two pre-registered controlled rounds, 60 compounds between them — against 77.6% (104/134) in the headline main round. Wrong-composition answers score as misses (~one main-round compound in five), so this inflates nothing; the main-round arm is the weakest-constrained (`scripts/analyze_misses.py`).

0.4 4. How well do LLMs elucidate real structures?

0.4.1 4.1 Headline performance

Solver agents work blind from formula + IR + ^{1}H + ^{13}C, returning up to three ranked candidates in bounded contexts reset every 2–12 compounds rather than one long context — the variable §4.3 shows what matters. The benchmark comprises all 194 compounds (134 spectrally-validated + 60 controlled-round; `data/benchmark_main/raw/`).

Table 2. Headline elucidation performance on IRSpectra-Bench (n=194). Overall and by difficulty stratum, with bootstrap 95% CIs.

metric	overall (n=194)	simple (n=98)	complex (n=96)
top-1 exact constitution	**28.4%** [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto ≥0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

A strictly-validated subset (134 main-clean + 57 controlled-clean = 191) reproduces this within ≈1 point on every metric — top-1 28.8%, 95% CI 23–36; recall 34.0%; simple 49.0% / complex 8.4% — so the asymmetric inclusion of pre-registered controlled sets does not drive it. The deliberate 50/50 difficulty balance does: reweighted to the 17.5%-simple composition of the eligible corpus, top-1 is 15.2% [10.6–20.4] and recall 19.8% [14.4–25.8] (§3). Per-stratum figures are unchanged; the reweighted number is what to read as performance on an arbitrary paper. Accuracy falls monotonically with size (Fig. 2): 60.5% top-1 at ≤15 heavy atoms, 28.3% at 16–25, 7.0% above 25 (Fig. S3).

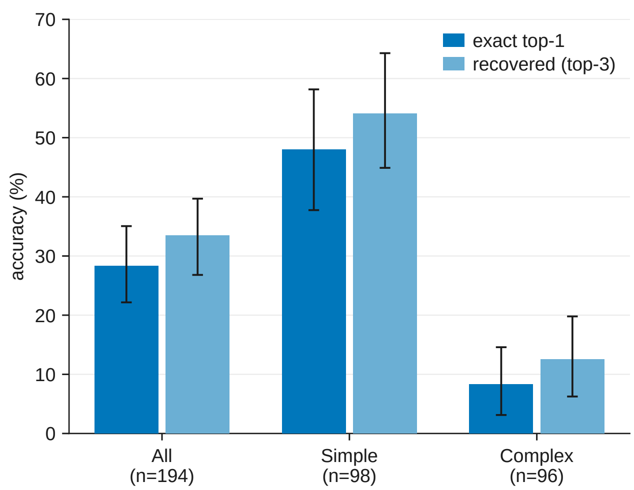

Fig. 2. Top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194), with bootstrap 95% confidence intervals. The benchmark separates a realistic difficulty range: simple targets are solved four to six times as often as complex ones on both metrics.

Of the 137 analysable top-1 misses (139 in all; two predictions did not parse; `scripts/analyze_misses.py`), 76.6% are constitutional isomers of the true structure against 23.4% with the wrong formula — not mostly regiochemistry: only 22.6% share the true Murcko scaffold, 2.9% reach Tanimoto $\geq$0.85, and the median isomeric-miss Tanimoto is 0.39. Strict regiochemistry accounts for a fifth of failures; most misses are larger rearrangements of the same atoms. Near-degenerate ^{1}H/^{13}C shifts under-determine connectivity; the misses show *wrong connectivity at the right composition.*

0.4.2 4.2 Reconciling with prior reports

Our 28% top-1 sits far below ≈100% on curated NMR-only demos[2]. Four choices inflate prior reports — difficulty, lenient top-3 scoring, starting-material hints, hand curation — and we name them without apportioning the gap. Trained baselines on simulated or NIST gas-phase IR reach 48–94% in-distribution[1,6,18]; ours is 28.4% on blind literature spectra. On ≤15-heavy-atom compounds our 60.5% approaches Alberts' 63.8% (IR-only, single instrument), but settings differ on every axis — modality, spectrum realism, library homogeneity — so neither number bounds the other. High in-distribution scores overstate real-world performance; the ≈40× MolPuzzle swing for one model (§1.1) bounds what unaudited near-100% claims can bear.

0.4.3 4.3 Methodology dominates: a within-compound control

The same 20 molecules were solved two ways: (a) one long context, no tools; (b) four agents, five compounds each, with RDKit formula-checking. Recovered rose 5% → 15% and top-1 0% → 15% (McNemar p=0.25 at n=20). Bounded, frequently-reset contexts with tool access may raise measured performance — a directional demonstration, not a quantified contribution to the headline gap.

0.4.4 4.4 Model comparison: the benchmark orders capability but separates only the extremes

We solved a fixed 24-compound subset blind with four Claude models spanning a wide capability range, including the newest (Fable 5), under one prompt, one scorer and one candidate budget (Fig. 3, Table 3).

One protocol asymmetry must be disclosed: the three comparison models solved the 24 compounds as four six-compound contexts each, whereas the Opus column reuses the headline run, where those items sat in one six-compound and two twelve-compound contexts — the variable that §4.3 measures at 5% → 15%. The Opus estimate is therefore not protocol-matched and, if anything, handicapped by the longer contexts; this touches neither the nesting nor the Fable-vs-Haiku contrast, and a clean re-run is outstanding in `docs/MODELS.md`.

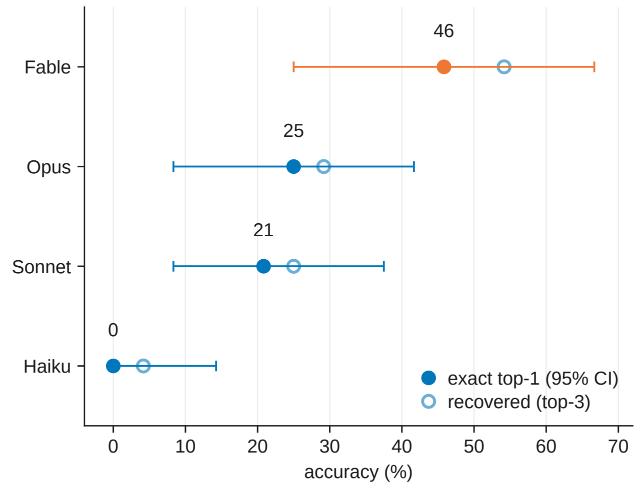

Fig. 3. Four-model comparison on a fixed 24-compound subset, solved blind under one protocol. Outcomes are strictly nested — each stronger model solves a superset of the weaker one's compounds — so the benchmark is capability-sensitive, but at n=24 it is underpowered to separate adjacent models (§4.4).

Table 3. Four-model comparison on a fixed 24-compound subset. All four models ran the identical blind protocol.

model	top-1	recovered	top-1 95% CI
Claude Fable 5	**46%**	54%	[25–67]
Claude Opus	25%	29%	[8–42]
Claude Sonnet	21%	25%	[8–38]
Claude Haiku	0%	4%	[0–14]

CIs are bootstrap except Haiku's: Clopper–Pearson exact for 0/24, where the percentile bootstrap is degenerate.

Outcomes are strictly nested (Haiku $\subset$ Sonnet $\subset$ Opus $\subset$ Fable); only Fable-vs-Haiku reaches significance (Holm p=0.006). Opus headline runs used longer contexts than the comparison models (§4.3).

0.4.5 4.5 Domain case study: battery-electrolyte chemistry

IRSpectra-Bench-Electrolyte (46 scored compounds across six electrolyte functional classes, held out and solved blind) matches the headline regime: top-1 26%, recovered 28%; the true structure enters the pool for 13/46 and ranks first in 12/13 — recall-bound, not domain-specific (Fig. S4, Table 4). Per-class intervals overlap (χ^2 p≈0.56); sp^3-C–F reaches 50%, sulfonyl and nitrile 12%.

Table 4. Per-class performance on IRSpectra-Bench-Electrolyte (n=46).

electrolyte class	n	top-1	95% CI	recovered (top-3)	95% CI
sp^3-C–F	8	**50%**	[22–78]	50%	[22–78]
carbonate	7	29%	[8–64]	43%	[16–75]
phosphoryl	8	25%	[7–59]	25%	[7–59]
glyme / oligoether	7	29%	[8–64]	29%	[8–64]
sulfonyl / sulfonate	8	12%	[2–47]	12%	[2–47]
nitrile	8	12%	[2–47]	12%	[2–47]

Intervals overlap throughout (six-class χ^2 $p \approx 0.56$; best-vs-worst Fisher exact $p \approx 0.28$), so the ordering is hypothesis-generating, not established — C–F couplings pin sp^3-C–F regiochemistry, while sulfonyl and nitrile targets turn on oxidation-state ambiguity and heteroaromatic substitution that $^1H/^{13}C$ shifts underdetermine. It is the recall-bound failure of §4.1 in a domain-specific guise, and it is where §5's *forward-verification* recipe plays a role *analogous* to (not a replacement for) computational-NMR validation.

0.4.6 4.6 Is the model reading the spectra? A formula-only control

We reran the blind protocol with spectra masked, leaving only the formula ([43]).

Table 5. Formula-only control. Paired on the same 60 compounds as §5.

condition	top-1	recovered (top-3)
formula only	**3/60 (5%)**	3/60 (5%)
formula + IR + ^1H + ^{13}C	14/60 (23%)	19/60 (32%)

Outcomes are perfectly nested (McNemar $p=0.001$). Accuracy is flat in source-paper year across all 194 ($r=-0.007$; Fig. 4), bounding pretraining recall. We claim a strong bound, not exclusion (§8).

0.4.7 4.7 Does the diagnosis hold outside one vendor? A four-vendor replication

Grok 4.6, Gemini 3.7 Flash and GPT-5.6 Sol solved the same 60-compound arm under the identical protocol (Table 6). Verification precision exceeds generation recall in every arm (Fig. 5); bootstrapping separates the paired gap for Claude (+52.5 points), GPT-5.6 Sol (+26.3) and Gemini (+23.3); Grok's gap (+9.2) is directional. Three models beat Claude on recall; candidate budgets differ (ours 2.20 vs 3.00 per compound), so recall rankings are approximate. A clean-clone control for Grok (recall 28/60 vs 32/60, McNemar $p=0.39$) bounds key leakage. Models ran through a coding-assistant harness with undisclosed reasoning tiers (§8).

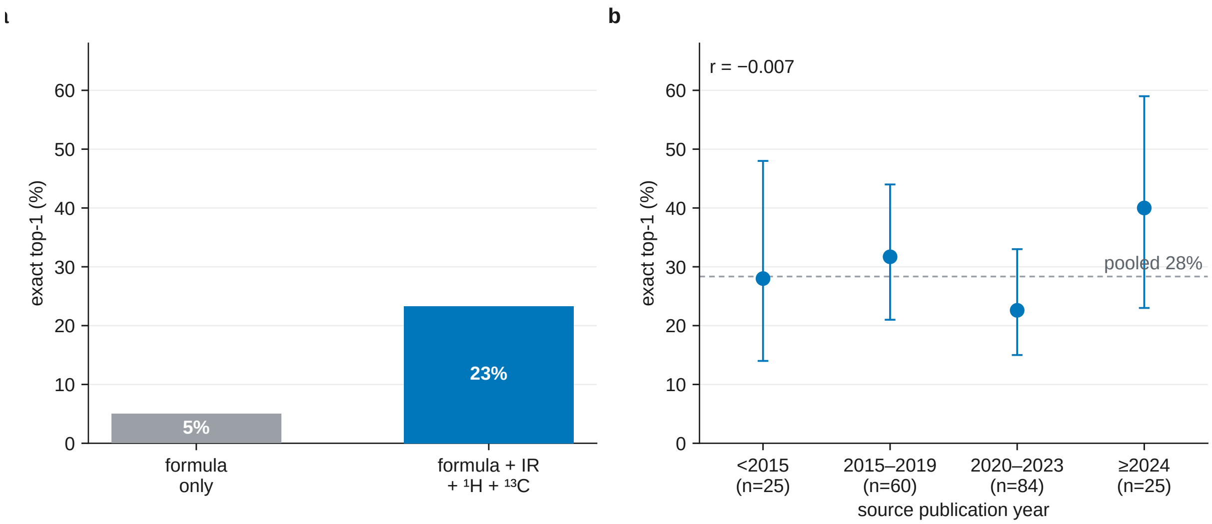

Fig. 4. Two contamination controls. (**a**) Formula-only vs full modality on 60 compounds. (**b**) Accuracy vs source-paper year (n=194).

Table 6. Cross-vendor decomposition on the 60-compound arm. Recall and precision have different denominators, so the criterion is the inequality rather than a difference.

model	generation recall	verification precision | recall	multi-candidate only
Claude Opus (Table 7)	19/60 = 32% [21–44]	16/19 = 84% [62–94]	10/13 = 77%
Grok 4.6	32/60 = 53% [41–65]	20/32 = 62% [45–77]	20/32 = 62%
Gemini 3.7 Flash	30/60 = 50% [38–62]	22/30 = 73% [56–86]	22/30 = 73%
GPT-5.6 Sol	25/60 = 42% [30–54]	17/25 = 68% [48–83]	16/24 = 67%

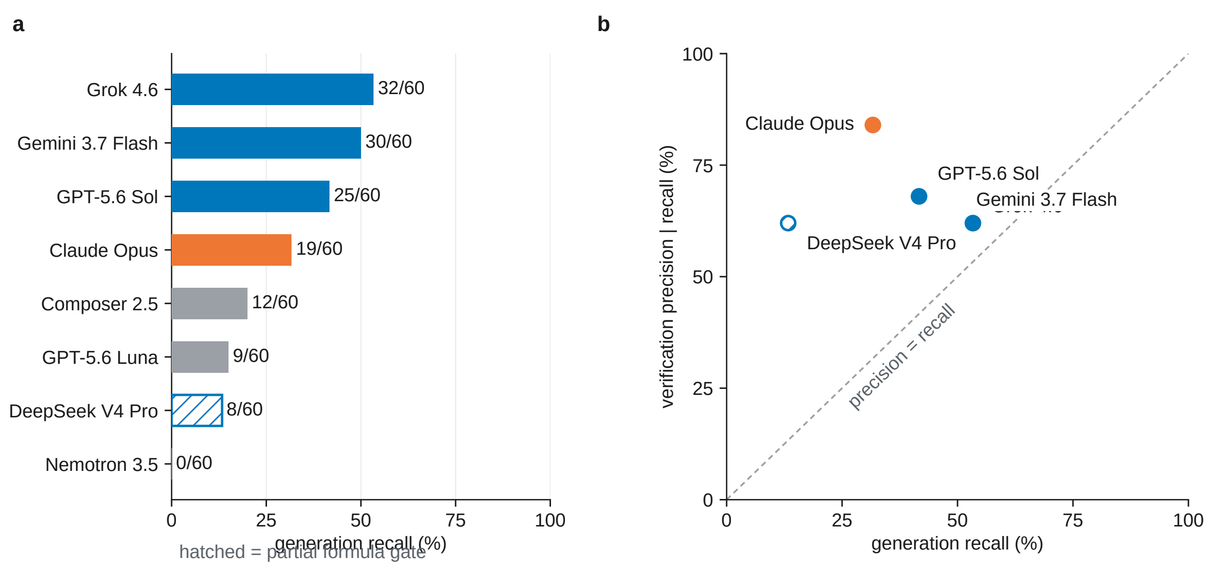

Fig. 5. Cross-vendor decomposition on the 60-compound arm: generation recall vs verification precision (§4.7).

0.5 5. Forward-verification elucidation

0.5.1 5.1 Method

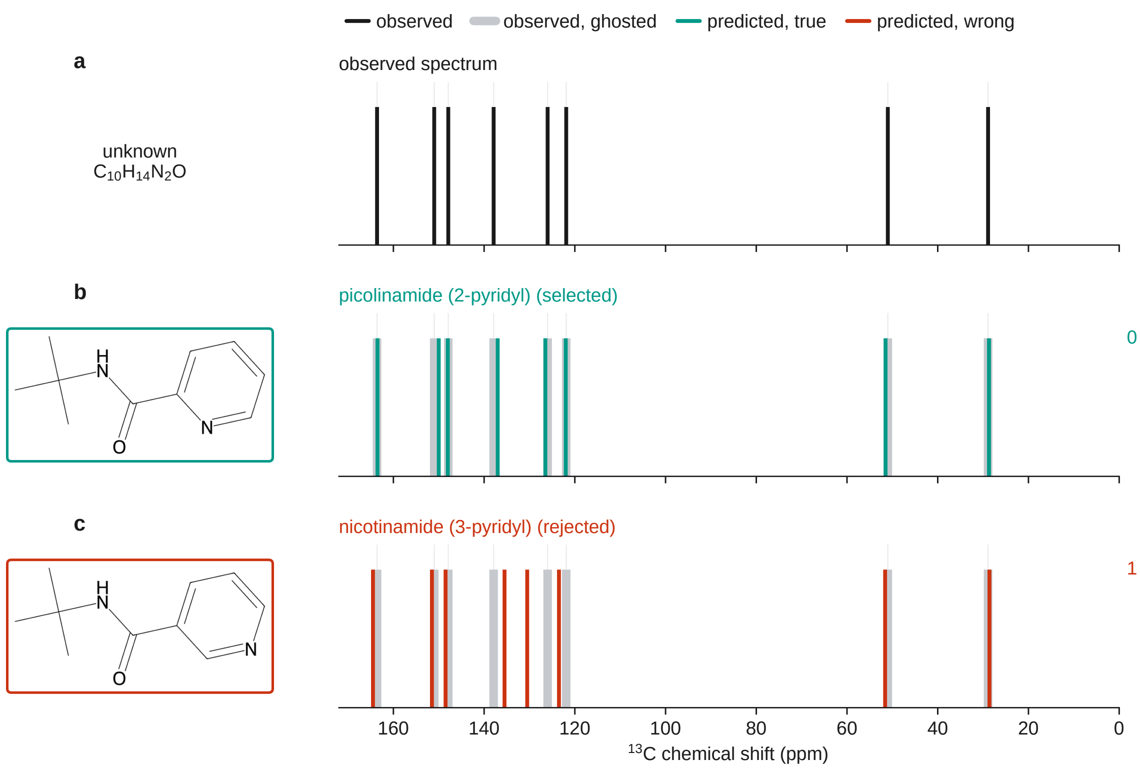

Fig. 6. Forward-verification on a benchmark regioisomer pair: picolinamide and nicotinamide are indistinguishable to the inverse task, but forward-predicted ^{13}C sticks separate the true isomer from the alternative (§5.1).

Sharper predictors (GNN, DFT-coupled models[44;45,46]) would move the verification ceiling, not search.

0.5.2 5.2 Result

The first column below is the 60-compound arm (v3 + v2-control) on which §5.3–§5.5 build; the second, the full benchmark. All 373 candidates were forward-predicted — nothing here is a lower bound; the self-ranking row re-derives Table 2 from §4's output.

Table 7. Forward-verification decomposition. Original arm and full benchmark.

	60-compound arm	full benchmark (n=194)
generation recall	19/60 (32%)	65/194 (34%)
top-1, solver self-ranking	14/60 (23%)	55/194 (28%)
top-1, forward-verified re-ranking	16/60 (27%)	58/194 (30%)
conditional on recall — self-ranking	14/19 (74%)	55/65 (85%)
conditional on recall — forward-verification	16/19 (84%)	58/65 (**89%**)
…single-candidate compounds within that set	6/19	28/65
multi-candidate only — self-ranking	8/13 (62%)	27/37 (73%)
multi-candidate only — forward-verification	10/13 (77%)	30/37 (81%)

When the true structure is among the candidates, forward-verification selects it in 58 of 65 cases (89%). Of the 65, 28 had a single candidate — scored by construction by any ranker — so the pooled figure is not the verifier's margin over chance. Where a choice existed (n=37), forward-verification gets 30/37 (81%) against a 54.0% derangement floor (§5.5) — +27.1 points; self-ranking 27/37 (73%).

The margin over self-ranking remains small and unresolved: seven compounds gained, four lost, McNemar exact p=0.55. We do not claim it. Precision is high in absolute terms while recall binds: top-1 moves only 28% → 30% because the true structure was never proposed for 129 of 194 compounds, which no re-ranking can repair. The verifier converts recall almost completely on *simple* targets (50/53, 94%) and ties self-ranking on *complex* ones (8/12, 67%). Elucidation factorises into two near-independent levers: a strong verifier and a generator at 34% recall that is the wall (Fig. 1).

0.5.3 5.3 Generate-wide: testing the recipe

The decomposition implies *generate wide, verify by forward prediction* — a chemistry analog of self-consistency sampling[47]. Ten solver agents proposed up to six regiochemistry-aware candidates per compound, pooled with the originals and re-ranked.

Table 8. Generate-wide vs original. On the 60-compound arm only, since wide generation was run there; the "original" column is Table 7's first.

	original (60-compound arm)	generate-wide
generation recall (true structure among candidates)	32%	**42%**
forward-verified top-1	27%	30%
verification precision (conditional on recall)	84%	72%

Wide generation lifts recall 32% → 42% and top-1 23% → 30% (McNemar p=0.34; Fig. 7). Recall plateaus at 42% on polycyclic targets; verification precision falls 84% → 72% — the training-free ceiling. Roughly a third of misses need only regiochemistry around a correct scaffold; two thirds need a scaffold never proposed.

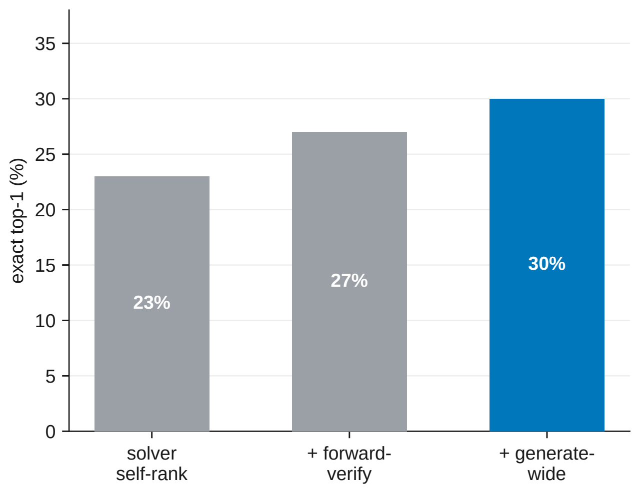

Fig. 7. Forward-verification inference ladder on the 60-compound arm (§5.3). Each stage adds a check on the same compounds; the hero bar is generate-wide top-1.

0.5.4 5.4 Non-LLM verifiers: a deterministic lookup and a learned model

§5.3 suggests replacing the verifier with a non-LLM ^{13}C predictor. We tested a HOSE-code[48]-style lookup and a small message-passing GNN, both trained on the same nmrshiftdb2 dump[49] and applied to the same §5.2 candidate sets so only the predictor changes. The GNN is deliberately modest; sharper purpose-built ^{13}C models exist[44,45,50].

Table 9. Verifier comparison, conditional on recall.

verifier	60-compound arm (n=19)	full benchmark (n=65)	held-out ^{13}C MAE
solver self-ranking	14/19 (74%)	55/65 (85%)	—
deterministic HOSE *lookup*	14/19 (74%)	55/65 (85%)	3.23 ppm
learned GNN (same data)	16/19 (84%)	**59/65 (91%)**	1.70 ppm
LLM forward-verifier (§5.2)	16/19 (84%)	58/65 (89%)	—

The HOSE lookup ties self-ranking; the GNN reaches 59/65 (91%) vs the LLM's 58/65 (89%), directionally (Table 9, Fig. S6). Wagen's MagNET-in-the-loop study[51–53] moves connectivity 46.4% → 55.4% on eight problems — same order as our ladder, underpowered at n=8.

0.5.5 5.5 Negative control

Deranging observed ^{13}C pairings collapses verification precision 89.2% → 73.8% mean (one-sided p=0.001); on multi-candidate sets alone, 81.1% vs a 54.0% floor (+27.1 points). Chamfer margin does not support selective prediction (flat top-1 across coverage fractions) — a reported null.

0.5.6 5.6 Is the recall wall task-intrinsic? A trained-generator probe

That ceiling is a property of training-free LLM elicitation, not the task. Pooling a small IRexp-fine-tuned ^{1}H/^{13}C → SMILES generator with Claude's candidates lifts recall 33.5% → 54.1% and top-1 28.4% → 35.1% (McNemar p=0.015; Fig. S5). Zero-shot without IRexp fine-tuning recovers 0/248; with it, 25%. NMR-Solver (52.9%)[27] and trained IR transformers (63.8%)[18] confirm the task is not bound at 28–30%.

0.6 6. The decomposition across the published literature

The split measured above applies to any system reporting top-1 = a and top-k = b: recall $\geq b$, conditional precision $\leq a/b$. Table 10 applies this to every system we could read in full (`scripts/literature_decomposition.py`).

Table 10. Recall and ranking loss from published top-k figures. Rows are comparable only within a correctness criterion; stereochemistry-strict and connectivity figures differ by 21 points on NMR-Solver's identical predictions. Candidate budgets are given because recall over three candidates is mechanically smaller than over ten.

system	data (n)	top-1	top-k (k)	recall	prec.
scored on connectivity					
this work, solver alone	literature (194)	28.4%	33.5% (3)	33.5%	84.6%
this work, + forward-verification	literature (194)	29.9%	33.5% (3)	33.5%	**89.2%**
NMR-Solver[27]	literature (450)	52.9%	67.3% (10)	≥67.3%	≤78.6%
NMRAgent[28]	literature (450)	61.6%	70.0% (10)	≥70.0%	≤88.0%
scored with stereochemistry					
this work	literature (194)	21.1%	25.8% (3)	≥25.8%	≤81.8%
NMR-Solver[27]	simulated (1,000)	66.9%	89.9% (10)	≥89.9%	≤74.4%
NMR-Solver[27]	literature (450)	31.6%	53.8% (10)	≥53.8%	≤58.7%
Espejo Morales[30]	education (236)	80.9%	90.0% (5)	≥90.0%	≤89.9%
Espejo Morales[30]	industrial (34)	20.6%	29.1% (5)	≥29.1%	≤70.9%
exact match, stereo handling not stated					
Alberts, 6–13 atoms[18]	NIST IR (3,455)	63.8%	84.0% (10)	84.0%	75.9%
Alberts, 5–35 atoms[18]	NIST IR (5,024)	59.9%	78.5% (10)	78.5%	76.4%
SpecX, random split[20]	simulated (99,439)	59.0%	81.8% (10)	81.8%	72.2%
SpecX, scaffold split[20]	simulated (99,439)	29.7%	50.6% (10)	50.6%	58.7%
IR-Agent[15]	NIST IR (905)	10.3%	21.6% (10)	21.6%	47.7%
criterion not stated					
Priessner[16]	experimental (34)	20.6%	—	26.5%	77.8%

Three groups changed only the data and recall carried 68% (NMR-Solver simulated → real), 70% (SpecX random → scaffold) and 83% (Espejo education → industrial) of each collapse. Published gains come mainly from candidate supply (NMRAgent ablation[28], NMRGym formula ablation[19]). Recall binds where spectra are real and heterogeneous (87–91% of our loss); ranking dominates on simulated or single-library data (38–39%). The field almost never reports whether the true structure was proposed at all ([16] excepted).

0.7 7. Discussion

Frontier LLMs are good verifiers (89% conditional precision) and weak proposers (34% recall) on real literature spectra. The contribution is a diagnosis with a bounded, training-free improvement, not a solved elucidator.

The split held across four Claude models, a battery-electrolyte subset, four verifiers and four vendor families (§6). Concurrent work sharpens it: Kliuev *et al.* place the best LLM at 24% on 105 peak lists[29]; Espejo Morales *et al.* show environment design moves proposal (71.1% on 15 molecules) while their education → industrial drop (80.9% → 20.6%) is our claim measured independently[30]; Wagen's MagNET loop[51,52] moves connectivity 46.4% → 55.4% on eight problems. Those three levers — proposal environment, external simulation, wider generation — attack one search problem.

Training is not foreclosed: IRexp fine-tuning lifts recall to 54% (§5.6). What compounds is open experimental data, honest benchmarks, and inference scaffolding that rides each new model. Two practical findings: bounded contexts with tool access may help (§4.3); use forward-predicted ^{13}C agreement to re-rank, not as an abstention gauge (§5.5). Learned embeddings rarely beat ECFP under controlled evaluation[54].

0.8 8. Limitations

Scoring is mechanical; solver runs were transcript-audited for zero ground-truth access (transcripts on request). **(i) Pretraining contamination** is bounded by formula-only (23% → 5%) and flat recency controls (§4.6) but not excluded; verbatim spectral strings in prompts remain a retrieval confound. **(ii) Human audit** of elucidation outputs is prepared (`data/audit/`) but not yet run. **(iii) Headline n=194 is Claude Opus**; cross-vendor evidence is on 60 compounds (§4.7). **(iv) Battery subset** uses literature electrolyte chemistry, not operando spectra. **(v) Constitution-only scoring** (21.1% with stereochemistry). **(vi) Single-sample scoring** per compound. **(vii) Organic literature bias. (viii) Author-transcribed peak lists**, not raw instrument files — a different regime from Espejo *et al.*[30].

0.9 9. Methods

Mining and resolution. PMC-OA full text was fetched from `s3://pmc-oa-opendata`, parsed deterministically, and resolved with OPSIN, RDKit and SELFIES, with an optional cached PubChem fallback.

Benchmark and agents. Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis, and de-duplicated across rounds by InChIKey. A compound was admitted only if its raw ^{1}H payload — multiplicities and J values as printed — could be re-extracted verbatim from the PMC-OA full text of its source article (`benchmark_v2.raw_1h_for`), which puts genuine coupling information in the prompt and, unavoidably, the source article's exact string (§8). The battery-electrolyte subset (§4.5) was drawn by SMARTS filters for six electrolyte functional classes (carbonate, sulfonyl/sulfonate, nitrile, sp^3-C–F, phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding compounds used elsewhere; it was J-enriched and spectrally validated. Solver and forward-prediction agents were independent Claude-Opus sub-agents under the same subscription, instructed closed-book and audited for zero web/answer access. Scoring used RDKit InChIKey-connectivity match; similarity Morgan(2, 2048) Tanimoto; forward-verification a symmetric chamfer distance over ^{13}C peak sets. The core protocol trains no model and uses no paid API; the §5.6 generator and the §5.4 learned ^{13}C verifier are the only trained components, reported as complements and fenced from the headline results.

Models and versions. Headline runs used Claude Opus via consumer subscription (2026-06-09–11); formula-only control 2026-07-28; forward-prediction extension 2026-08-07. No model snapshot or decoding parameters can be reported — the harness exposes neither (§9). Fixed are frozen per-compound outputs and mechanical scorers (`docs/MODELS.md`).

Reproducibility. Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions and scorer outputs are released; the sampler, scorer and forward-verification harness are scripted end-to-end.

0.10 Supporting Information figures

These are supplied as a separate Electronic Supplementary Information document (`docs/paper_esi.pdf`, built by `scripts/build_pdf.py`); they are listed here for reference.

- **Fig. S1** (`docs/figures/fig0_overview.png`) — study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free core pipeline.
- **Fig. S2** (`docs/figures/fig4_dataset.png`) — IRexp composition: IR records → NMR-paired → structure-linked → full IR + ^{1}H + ^{13}C + structure quadruples.
- **Fig. S3** (`docs/figures/fig2_size.png`) — accuracy vs molecular size (heavy-atom bucket); the monotonic 60% → 7% top-1 gradient.
- **Fig. S4** (`docs/figures/fig6_electrolyte.png`) — top-1 and recovered accuracy on IRSpectra-Bench-Electrolyte by battery-electrolyte chemical class (n=46): sp^3-C–F easiest (50%), sulfonyl and nitrile hardest (12%); overall 26%/28%.
- **Fig. S5** (`docs/figures/fig_generator_probe.png`) — trained-generator probe (§5.6; complement, not training-free protocol): generation recall and deterministic-HOSE top-1 on the 194-compound benchmark for Claude / + scaffold enumeration / + trained generator. Enumeration's near-degenerate isomers collapse the verifier (28.4 → 16.0%) while the generator's formula-correct candidates convert (28.4 → 35.1%).
- **Fig. S6** (`docs/figures/fig_verifier.png`) — learned-verifier probe (§5.4; complement, not training-free protocol). (A) Conditional-on-recall top-1 (n=65, whole benchmark) for four verifiers: a GNN trained on the same nmrshiftdb2 data as the HOSE lookup reaches the LLM verifier's level (91% vs 89%) where the lookup (85%) does not move off the solver's own ranking. (B) Held-out ^{13}C MAE — the learned model is ≈2× sharper (1.70 vs 3.23 ppm).

0.11 Author contributions

I.Y.: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing — original draft. **R.S.:** methodology, software, formal analysis, investigation, validation (trained-generator and learned-verifier probes, §5.4 and §5.6), writing — review and editing. **R.A.V.-H.:** conceptualization, methodology, supervision, writing — review and editing.

0.12 Conflicts of interest

There are no conflicts to declare.

0.13 Data availability

Table 11 lists every released component and the script that regenerates it, all of them in the project repository. The archival deposit — a complete frozen snapshot of dataset, benchmark, answer keys, predictions, scripts, figure regeneration and the expert-audit package — is deposited on Zenodo, and its DOI is supplied at proof stage; GitHub is the development mirror and the Zenodo record the citable version. IRexp is also mirrored for ML loaders at huggingface.co/datasets/ilkhamfy/IRexp; prefer `data/train_no_bench.jsonl.gz` there for fine-tuning without benchmark leakage.

Table 11. Released artefacts, and the data and code behind each one.

component	data and code
IRexp and its structure-complete split	`data/irexp/`, `data/irexp_resolved/` — `spectro_scraper/` (mining pipeline)
benchmark rounds and the within-compound control	`data/benchmark*/` — `scripts/benchmark_v2.py`
ground-truth integrity audit	`data/benchmark*/clean_qids.json` — `scripts/validate_benchmark.py`
headline accuracy (Table 2)	`scripts/score_main.py`
battery-electrolyte subset (§4.5)	`data/benchmark_electrolyte/` — `scripts/build_electrolyte_bench.py`, `scripts/score_electrolyte.py`
formula-only contamination control (§4.6)	`data/modality/` — `scripts/modality_ablation.py`
publication-recency control (§4.6)	`data/audit/recency_control.json` — `scripts/contamination_recency.py`
forward-verification, original arm (§5.2)	`data/fverify/` — `scripts/forward_verify.py`
…extended to all 194 compounds	`data/fverify_main/` — `scripts/forward_verify_main.py`, `scripts/forward_verify_all.py`
generate-wide arm (§5.3)	`data/gw/`, `data/fverify2/` — `scripts/score_generate_wide.py`, `scripts/ladder_significance.py`
…its coverage gap, closed	`data/fverify_gw/` — `scripts/forward_verify_gw.py`
non-LLM verifier comparison (Table 9)	`data/fverify/hose_results.txt`, `data/fverify/verifier_table_results.txt` — `scripts/hose_predict.py` (incl. coverage), `scripts/verifier_table.py`, `scripts/verifier_leakage.py`
negative control and selective prediction (§5.5)	`scripts/verifier_diagnostics.py`
what a miss is, and isomer separability (§4.1, §5.1)	`scripts/analyze_misses.py`, `scripts/isomer_separability.py`

component	data and code
difficulty-threshold sensitivity (§3)	`scripts/difficulty_sensitivity.py`
licence-pool split (§2.1)	`scripts/split_license_pools.py`
recall headroom and scaffold enumeration	`scripts/analyze_recall_headroom.py`, `scripts/enumerate_isomers.py`, `scripts/closing_the_gap.py`
blinded expert-audit package (§8)	`data/audit/` — `scripts/make_audit_sample.py`
corpus reweighting (§3)	`scripts/corpus_reweight.py`
cross-vendor recall at matched budget, and the paired gap (§4.7)	`data/cross_vendor/` — `scripts/cross_vendor_budget.py`, `scripts/cross_vendor_gap.py`
ring-system names in peak assignments (§3)	`scripts/prompt_leakage.py`
paraphrase-invariant benchmark, for the control not yet run (§8)	— (regenerated, not deposited) — `scripts/jitter_control.py`
manuscript integrity gates	`scripts/check_manuscript.py`, `scripts/verify_statistics.py`, `scripts/check_compression.py`, `scripts/check_layout.py`

Trained complements, both fenced from the training-free protocol. The §5.6 generator ships as a self-contained bundle at `contrib/generator_probe/` — candidates, the de-leaked split with its InChIKey-14 manifest, the verification scripts (`scripts/closing_the_gap_gen.py`, `scripts/forward_verify_gen.py`, `scripts/verify_leakage_exact40.py`) and the blind forward-prediction outputs behind its verified top-1 (`data/fverify_gen/`), so its scorer runs with no missing predictions. The §5.4 learned verifier ships `scripts/gnn_predict.py` (extract / train / score / control), the trained model `data/nmrshiftdb/gnn_c13.pt`, per-compound results and both leakage checks (`data/fverify/gnn_results.txt`), and the write-up `docs/VERIFIER_PROBE.md`. Model checkpoints are deposited on Zenodo. Both trained predictors of §5.4 need the nmrshiftdb2 dump, which we cannot redistribute; the README gives the one-line fetch.

Companion technical notes: `docs/BENCHMARK.md`, `docs/FORWARD_VERIFY.md`, `docs/MODALITY_ABLATION.md`, `docs/EXPERT_AUDIT_PROTOCOL.md`, `docs/MODELS.md`.

Two cautions for re-users. The public `irexp_release/train` split does not hold out IRSpectra-Bench — it overlaps by 117/200 InChIKey-14 — so de-leak with `contrib/generator_probe/build_exp_manifest.py` before training anything that will be evaluated on the benchmark. And the held-out answer keys `data/audit/key.jsonl` and `data/modality/key.json` are deposited but flagged *withhold from blinded reviewers*.

0.13.1 Licensing and attribution

IRexp is derived from two open-access source pools and redistributed under terms compatible with each (see `data/NOTICE`). **(a) Redistribution:** we release only *extracted numeric data* — IR band lists, $^{1}H/^{13}C$ shift lists, and resolved structures (SMILES/SELFIES/InChIKey) — plus each record's source DOI/accession, not source full text, figures, or PDFs. **(b) Two separable pools:** 119,345 records derive from the PMC Open-Access Subset (CC-BY-4.0) and 1,888 from the Chemotion RADAR4Chem FT-IR deposit (CC-BY-SA-4.0). The two are separable losslessly by `source_doi` — Chemotion records carry the `10.22000` prefix — and `scripts/split_license_pools.py` materialises them as two files with each record stamped `license`, so users may take the CC-BY pool alone; any combined or Chemotion-derived release carries CC-BY-SA-4.0 to honour the ShareAlike term. Code is released under the MIT License. **(c) Attribution:** re-users must cite this dataset (Zenodo DOI above) and attribute the original publications via each record's `source_doi`.

0.14 Acknowledgements

0.14.1 Use of AI tools

This work studies a large language model, and LLMs were also used as instruments and as writing aids; we state both roles explicitly. **As an object of study:** all reported elucidation, forward-prediction and verification results were produced by Claude models invoked under the protocol of §3 and §9 — these are the measurements the paper reports, not assistance in producing it. **As a writing aid:** the authors used an LLM-based coding assistant for figure generation, analysis scripting, manuscript copy-editing, and for the internal review pass that produced several of the corrections recorded in the repository history. No text, number, figure or citation in this manuscript was accepted without author verification against the released data and code; every quantitative claim regenerates from the scripts in `scripts/`. The authors take full responsibility for the content.

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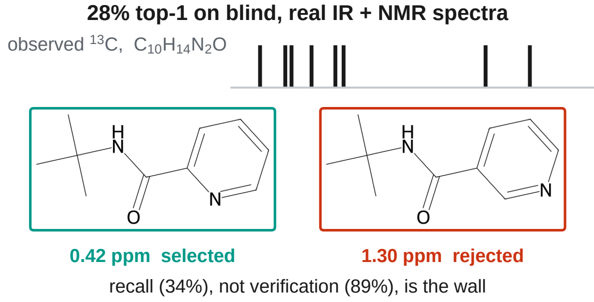

IRSpectra-Bench and IRexp on real literature IR + ^{1}H + ^{13}C: candidate recall, not verification, limits LLM elucidation — the true structure is proposed for 34% of 194 compounds and, once proposed, selected 89% of the time, lifting top-1 from 28% to 30%.